

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/2

PAPER 2 Theory

NOVEMBER 2019 SESSION

2 hours

Candidates answer on the question paper

Additional materials: Calculator (Optional)

Allow candidates 5 minutes to count pages before the examination.

This booklet should not be punched or stapled and pages should not be removed.

The Periodic Table is provided on page 20.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Write your centre and candidate number in the boxes on the top right corner of every page of this paper.

Check if the booklet has all the pages and ask the invigilator for a replacement if there are duplicate or missing pages.

Write your answers in the spaces provided on the question paper.

Section A

Answer **all** questions.

Section B

Answer any **two** questions.

Section C

Answer any **two** questions.

Section D

Answer any **two** questions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of 20 printed pages.

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[Turn over

Section A

Answer all the questions in the spaces provided.

1. (a) State any **two** differences between inhaled and exhaled air.

1
2
.....
.....

[2]

- (b) Transpiration is the loss of water through plant leaves.

- (i) State any **one** advantage of transpiration to the plant.

.....
.....

[1]

- (ii) State **one** disadvantage of excessive transpiration.

.....
.....

[1]

- (iii) State any **two** factors which increase the rate of transpiration.

1
2
.....
.....

[2]

2. (a) Describe a natural ecosystem.

.....

[2]

- (b) (i) Define the term *balanced diet*.

.....

[2]

- (ii) Describe the importance of calcium to a pregnant woman.

.....

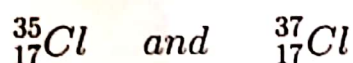
[2]

- (iii) State the advantage of eating liver.

.....

[1]

3. (a) Chlorine gas has two types of atoms as shown:



- (i) State the name given to the two types of the chlorine atoms.

.....

[1]

- (ii) Calculate the number of neutrons in ${}_{17}^{35}\text{Cl}$.

.....

[1]

- (b) Chlorine reacts with sodium to form sodium chloride, NaCl .

- (i) Name the type of bonding in sodium chloride.

[1]

- (ii) Draw a dot and cross diagram to show the bonding in sodium chloride.

[2]

- c) State any **two** physical properties of sodium chloride.

1
2

[2]

4. (a) Indigestion is caused by too much dilute hydrochloric acid in the stomach. It is cured by ingesting anti-acid tablets.

State, with a reason, the acid-base nature of the chemical present in the anti-acid tablets.

acid-base nature
reason
.....

[2]

- (b) (i) Iron is extracted from an iron compound found in haematite.
Name the iron compound in haematite.

[1]

- (c) Two other solid raw materials are fed into the blast furnace together with haematite. Name the **two** raw materials and state a function for each of these materials.

raw material

function

raw material

function

[4]

5. (a) Fig.5.1 shows a stroke in the operation of an engine.

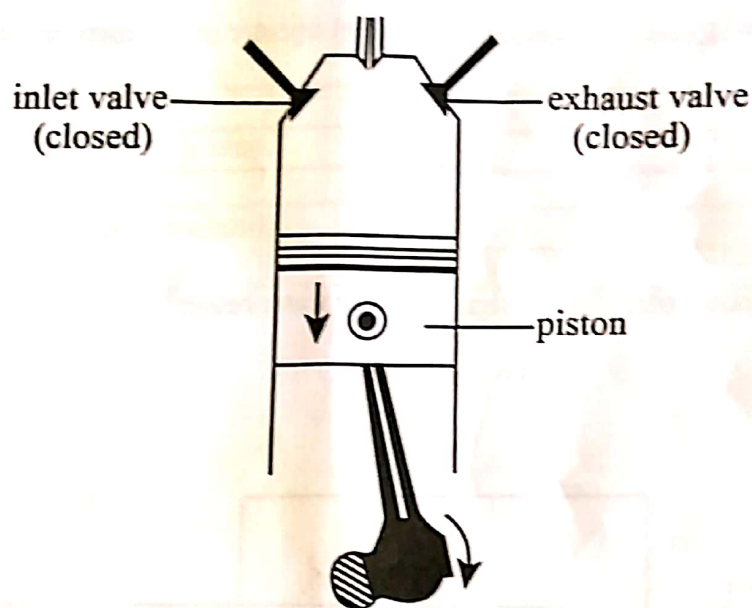


Fig.5.1

- (i) Identify, giving **two** reasons, the stroke shown.

stroke

reasons: 1

2

.....

[3]

- (b) State the role of a fuel injector in a petrol engine.

.....

.....

.....

[1]

- (ii) State the role of a carburettor.

.....

.....

[1]

- (iii) Explain the advantage of a fuel injector over a carburettor.

.....

.....

.....

[2]

6. Fig.6.1 shows part of the design of a solar water heater.

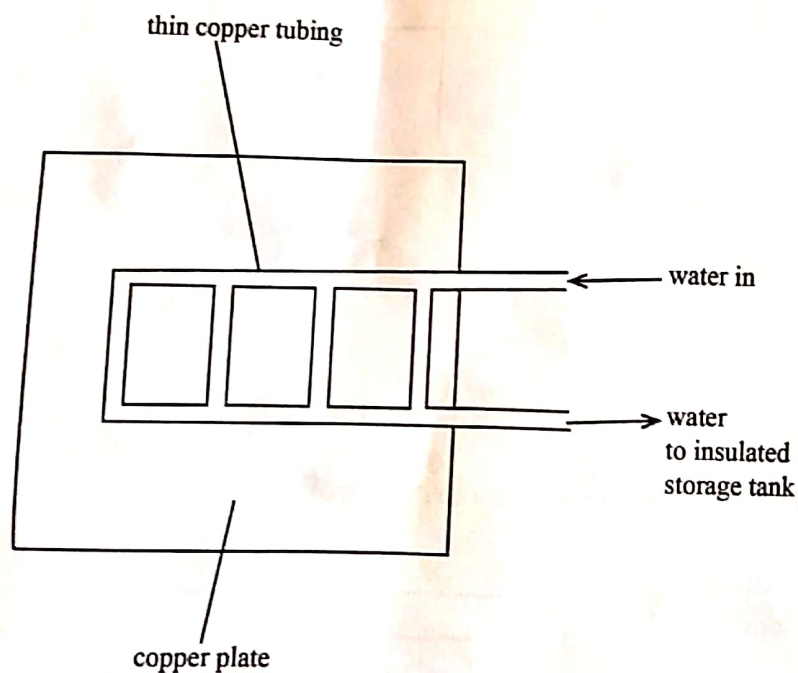


Fig.6.1

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- (a) State, with a reason, the most suitable place for placing the solar water heater for best results.

place
reason
.....

[2]

- (b) State, with a reason, the paint colour on the copper plate.

colour
reason

[2]

- (c) Explain why

(i) a thin copper tubing is used,

(ii) the storage tank is insulated.
.....

[2]

Section B

Answer any two questions in the spaces provided.

7. (a) Fig.7.1 shows a sketch diagram to represent double circulation in mammals.

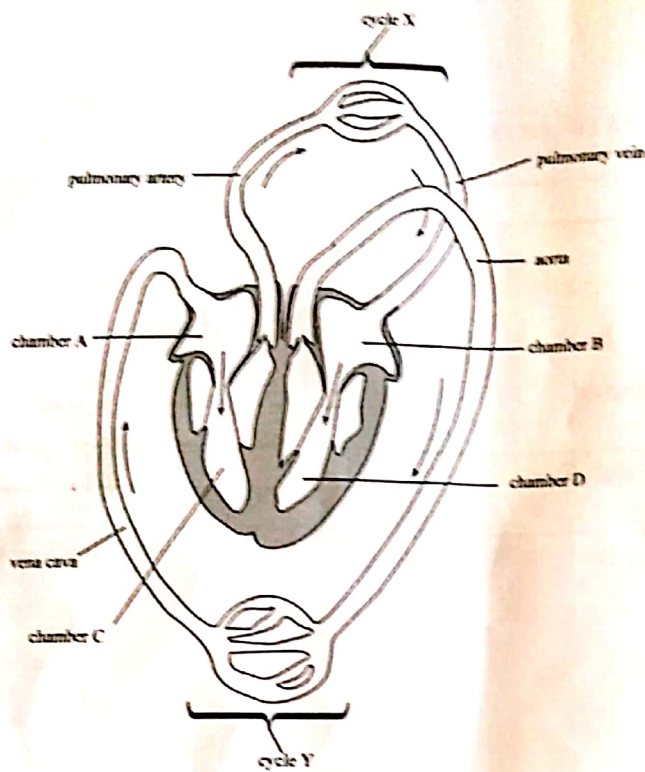


Fig.7.1

- (i) Deduce the types of circulation represented by cycles X and Y.

Cycle X

cycle Y

[2]

- (ii) Suggest the reason for differences in the thickness of the walls of chambers C and D.

[2]

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(iii) State any **three** symptoms of malaria.

- 1 high
- 2 fever
- 3 chills

[3]

(iv) State a symptom of ebola which is different from symptoms of malaria.

swollen lymph nodes

[1]

(v) State any **two** effects of inhaling glue.

- 1 respiratory problems
- 2 brain damage

[2]

8. (a) Fig.8.1 shows a child suffering from a deficiency disease.



Fig.8.1

(i) Name the deficiency disease which the child is suffering from.

.....

[1]

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(ii) Describe how the disease named in (i) could be prevented.

.....
.....

[2]

(b) Describe the route of the sperm from the testis to the oviduct.

.....
.....
.....

[4]

(c) State **one** advantage of using condoms during sexual intercourse.

.....

[1]

~~*~~ (d) Define the term *fertilisation*.

.....
.....

[2]

9. (a) Fig.9.1 shows gaseous exchange in the alveolus of a mammal.

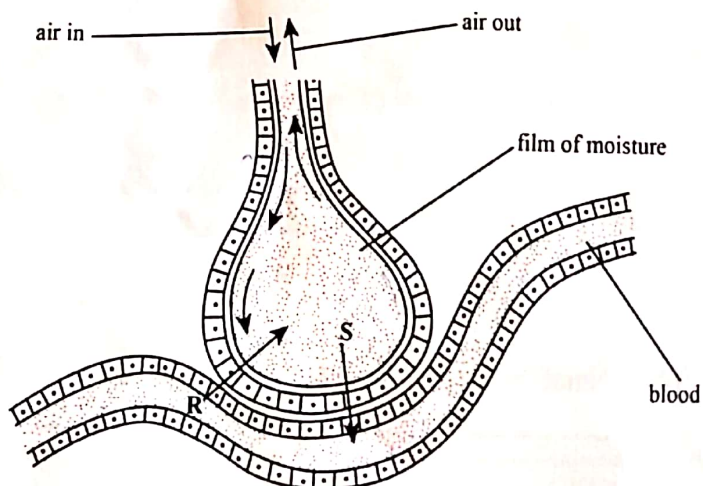


Fig.9.1

- (i) Name the gases moving in the directions shown by the arrows R and S.

R
S

[2]

- (ii) Describe and explain how the alveolus is adapted for gaseous exchange.

.....
.....
.....
.....

[4]

- ★(b) Define the terms *plasmolysis* and *turgidity*.

plasmolysis
.....
turgidity
.....

[4]

Section C

Answer any **two** questions in the spaces provided.

10. (a) (i) Define the term *atom*.

[1]

- (ii) State the **two** sub-atomic particles found in the nucleus of an atom.

1
2

[2]

- (b) Determine the empirical formula of a compound made up of 75% by mass carbon and 25% by mass hydrogen.

[4]

- (c) Sodium hydroxide solution reacts with dilute nitric acid to form a salt and water.

- (i) State the type of reaction that occurs.

[1]

- (ii) Determine the chemical formula of the salt.

[2]

11. (a) Outline the stages involved in the extraction of nitrogen from liquid air.

.....

.....

.....

.....

.....

[4]

- (b) Oxygen can be obtained from the electrolysis of acidified water.

- (i) Name the acid used to acidify the water.

.....

[1]

- (ii) Explain why the water is acidified.

.....

.....

[2]

- (iii) Explain why the volume of oxygen obtained during the electrolysis process is half that of hydrogen.

.....

.....

[2]

- (iv) State any **one** use of oxygen.

.....

[1]

12. Fig.12.1 shows the production of sulphuric acid by the contact process.

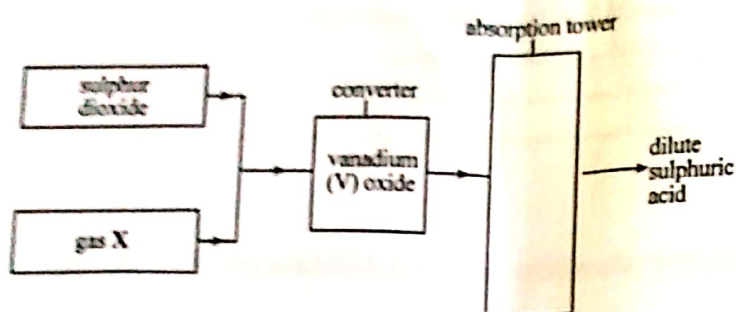


Fig.12.1

- (a) (i) Name gas X.

[1]

- (ii) State the role of vanadium (V) oxide.

[1]

- (iii) Explain why sulphur trioxide is **not** directly added to water.

[2]

- (iv) Define the terms *exothermic* and *reversible*.

exothermic

reversible

[2]

- (v) Name the substance which is formed in the absorption tower.

[1]

- (b) Ammonium sulphate, $(\text{NH}_4)_2\text{SO}_4$, is a fertilizer produced from sulphuric acid. Calculate the percentage composition by mass of nitrogen in ammonium sulphate.

[3]

Section D

Answer any two questions in the spaces provided.

13. (a) Fig.13.1 shows an alternating current (a.c) generator.

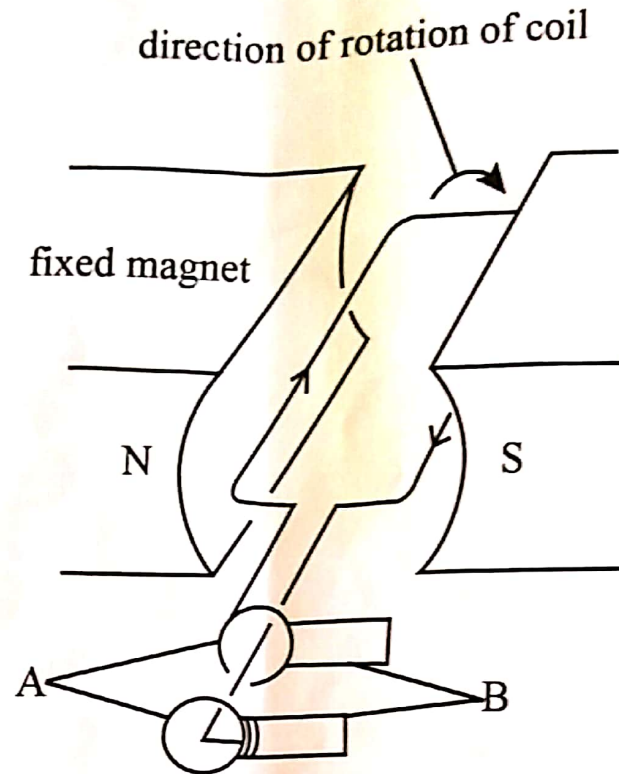


Fig.13.1

- (i) Name the parts labelled A and B.

A
B

[2]

- (ii) Describe how the a.c. generator produces electricity.

.....
.....
.....
.....

[4]

(iii) Sketch a graph of output voltage of the generator against time.

[2]

(b) Explain the effect of using stronger magnets on the magnitude of the output voltage.

.....
.....

[2]

14. (a) A gear system has ten teeth in the driving gear and thirty teeth in the driven gear.

(i) Calculate the velocity ratio, VR, of the gear system.

[2]

(ii) Determine the efficiency of the system if its mechanical advantage, MA, is 2.

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[2]

- (iii) Give any **two** reasons why the efficiency of a machine is always less than 100 %.

1
2

[2]

- (iv) State any **two** ways by which the efficiency of a machine can be improved.

1
2

[2]

- (b) State any **two** types of machines apart from gears.

.....
.....
.....

[2]

15. (a) Describe how electricity is generated at a thermal power station.

.....
.....
.....
.....

[4]

- (b) State any **two** disadvantages of using coal as a source of fuel for a thermal power station.

.....
.....

[2]

- (c) Give the main difference between a thermal power station and a hydroelectric power station.

.....
.....

[2]

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- (d) State the type of energy possessed by water which is in a dam.

[1]

- (e) State the Standard International (S.I) unit of energy.

[1]

DATA SHEET
The Periodic Table of the Elements

Group																	
I	II											III	IV	V	VI	VII	0
																	1 H Hydrogen
7 Li Lithium	9 Be Beryllium											11 B Boron	12 C Carbon	14 N Nitrogen	16 O Oxygen	18 F Fluorine	20 Ne Neon
23 Na Sodium	24 Mg Magnesium											27 Al Aluminium	28 Si Silicon	31 P Phosphorus	32 S Sulphur	35.5 Cl Chlorine	40 Ar Argon
39 K Potassium	40 Ca Calcium	45 Sc Scandium	46 Ti Titanium	51 V Vanadium	52 Cr Chromium	55 Mn Manganese	56 Fe Iron	59 Co Cobalt	58 Ni Nickel	64 Cu Copper	65 Zn Zinc	70 Ga Gallium	73 Ge Germanium	75 As Arsenic	79 Se Selenium	80 Br Bromine	84 Kr Krypton
85 Rb Rubidium	86 Sr Strontium	89 Y Yttrium	91 Zr Zirconium	93 Nb Niobium	96 Mo Molybdenum	101 Tc Technetium	102 Ru Ruthenium	103 Rh Rhodium	106 Pd Palladium	108 Ag Silver	112 Cd Cadmium	115 In Indium	119 Sn Tin	122 Sb Antimony	128 Te Tellurium	127 I Iodine	131 Xe Xenon
133 Cs Caesium	137 Ba Barium	139 La Lanthanum	178 Hf Hafnium	181 Ta Tantalum	184 W Tungsten	186 Re Rhenium	190 Os Osmium	192 Ir Iridium	195 Pt Platinum	197 Au Gold	201 Hg Mercury	204 Tl Thallium	209 Pb Lead	209 Bi Bismuth	210 Po Polonium	210 At Astatine	210 Rn Radon
87 Fr Francium	88 Ra Radium	89 Ac Actinium															

*58-71 Lanthanoid series
*90-103 Actinoid series

Key
a = relative atomic mass
X = atomic symbol
b = proton (atomic) Number

140 Ce Cerium	141 Pr Praseodymium	144 Nd Neodymium	150 Pm Promethium	152 Sm Samarium	157 Eu Europium	162 Gd Gadolinium	163 Tb Terbium	165 Dy Dysprosium	167 Ho Holmium	173 Er Erbium	175 Tm Thulium	177 Yb Ytterbium	175 Lu Lutetium
232 Th Thorium	232 Pa Protactinium	238 U Uranium	238 Np Neptunium	244 Pu Plutonium	244 Am Americium	247 Cm Curium	251 Bk Berkelium	259 Cf Californium	264 Es Einsteinium	267 Fm Fermium	271 Md Mendelevium	270 No Nobelium	271 Lr Lawrencium

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)